

Appliances Energy Consumption Prediction Project

W.M.Madushani Weerasekara
1st July 2025

Table of Contents

Introduction:	5
Data Insights:	7
Exploratory Data Analysis (EDA) Findings.....	7
Key Visualizations	9
Visualization 1 Temporal plot/line plot: Appliance Energy Consumption over Time	9
Visualization 2: Autocorrelation Plot of Appliance Energy Usage	10
Visualization 3: Boxplot for Outlier Detection in Appliance Energy Usage	11
Visualization 4: Correlation Heatmap of Features.....	13
Visualization 5: Histogram – Distribution of Appliance Energy Usage	14
Visualization 6: Scatter Plots – Selected Features vs Appliance Usage	16
Box Plot of Appliances Energy Usage (After Capping Outliers)	17
Box Plot after Capping Outliers.....	18
Outlier Treatment Decision	18
Preprocessing	19
1. Data Cleaning and Preparation.....	19
Outlier Detection and Treatment.....	20
3. Scaling and Normalization.....	20
Feature Engineering	21
1. Time-Based Features.....	21
2. Domain-Specific Features	21
3. Rolling and Lagged Features.....	21
4. Interaction Features	22
Feature Selection and Justification	22
Justification for Engineered Features:	22
Model Design:	23
Baseline Machine Learning Models.....	23
Models Used	23
Evaluation and Key Observations.....	23

Why These Baselines Matter	24
Deep Learning Model Architecture	25
Justification of Design Choices	26
Results:	27
Deep Learning Model Performance.....	27
Predicted vs. Actual values of LSTM.....	29
Model Optimization.....	30
Challenges and Solutions	32
Challenge: Achieving Satisfactory Model Accuracy	32
Solution: Hyperparameter Tuning with GridSearchCV and Optuna.....	32
Additional Solutions and Improvements.....	32
Conclusion	34

Table of Figures

Figure 1: Temporal plot/line plot: Appliance Energy Consumption over Time.....	9
Figure 2: Autocorrelation Plot of Appliance Energy Usage.....	10
Figure 3: Boxplot for Outlier Detection in Appliance Energy Usage	11
Figure 4: Correlation Heatmap of Features	13
Figure 5: Distribution of Appliance Energy Usage	14
Figure 6: Scatter Plots – Selected Features vs. Appliance Usage.....	16
Figure 7: Box Plot of Appliances Energy Usage (After Capping Outliers)	17
Figure 8: Predicted vs. Actual values of LSTM	29

List of Tables

Table 1: Deep Learning Model Performance.....	27
---	----

Introduction:

Accurately predicting energy consumption in residential and commercial environments is a critical challenge in today's data-driven world. As smart meters, sensor networks, and the Internet of Things (IoT) proliferate, vast amounts of granular energy usage data are being generated at unprecedented rates. Harnessing this data to build predictive models offers significant benefits: it enables consumers to make informed decisions, helps utility companies optimize load balancing and demand response, and ultimately supports the broader goals of sustainability and energy efficiency.

In particular, **appliance-level energy prediction** has emerged as a key area of interest within the broader field of energy analytics. Unlike aggregated building-level forecasting, appliance-level prediction enables more granular monitoring and control. This fine-grained perspective is invaluable for:

- **Home automation and smart scheduling:** Allowing devices to run at optimal times based on predicted usage.
- **Energy conservation and cost savings:** Empowering households and facilities to identify high-usage patterns and reduce unnecessary consumption.
- **Smart grid integration:** Facilitating dynamic pricing, load forecasting, and the efficient integration of renewable energy sources.

However, modeling appliance energy consumption presents significant technical challenges. The data is inherently **time-series** in nature, with strong dependencies on past usage, time of day, week, season, and external factors such as weather conditions and occupant behavior. Additionally, appliance usage can be highly variable and non-linear, featuring periodic spikes, sudden drops, and complex interactions between environmental variables. These characteristics demand modeling approaches that are both robust to noise and sensitive to temporal patterns.

This project leverages the **Appliance Energy Prediction Dataset**, which comprises timestamped records of appliance-level energy consumption along with a suite of contextual variables, including indoor and outdoor temperatures, humidity levels, atmospheric pressure, and more. The richness and complexity of this dataset make it an ideal testbed for both classical and modern predictive modeling techniques.

The **primary objective** of this project is to design, implement, and rigorously evaluate a suite of machine learning and deep learning models for accurate predicting of appliance energy usage. The approach is characterized by:

- **Thorough exploratory data analysis (EDA):** Identifying key patterns, anomalies, and relationships in the data.
- **Robust data preprocessing:** Including handling missing values, treating outliers, and scaling features for optimal model performance.
- **Domain-informed feature engineering:** Creating time-based, lagged, rolling, and interaction features, as well as encoding domain-specific effects such as holidays and seasons.
- **Model benchmarking:** Establishing baselines using classical regression models and then advancing to state-of-the-art deep learning architectures (LSTM, GRU, CNN-LSTM) tailored for time-series forecasting.
- **Advanced hyperparameter optimization:** Employing both GridSearchCV and Optuna to systematically fine-tune models for maximum predictive accuracy.
- **Comprehensive evaluation:** Using RMSE, MAE, and visualizations to compare models and assess generalization to unseen data.

The report is structured to reflect best practices in both industry and academic data science workflows. Each stage—from initial data exploration through to final model deployment—is justified and documented with a focus on reproducibility, interpretability, and actionable insights. By the end of this project, the reader will have a clear understanding of both the challenges and opportunities inherent in appliance-level energy prediction, and how modern data-driven techniques can deliver measurable improvements in forecasting accuracy.

This work not only demonstrates technical competence in applied machine learning and deep learning but also reflects a forward-thinking approach to the growing demands of sustainable energy management in an increasingly connected world.

Data Insights:

Exploratory Data Analysis (EDA) Findings

A thorough exploratory data analysis was conducted to understand the structure, relationships, and temporal patterns present in the Appliance Energy Prediction Dataset. The following are the key insights derived from the EDA, supported by visualizations and quantitative observations:

1. Data Quality and Structure

- The dataset is time-indexed, with records collected at regular intervals, and features a combination of target (Appliances energy consumption) and explanatory variables (temperatures, humidity levels, weather data, etc.).
- Missing values were minimal and primarily addressed using forward-fill imputation, ensuring data continuity for time series modeling.

2. Target Variable Distribution

- **Distribution Analysis:**
 - The target variable, appliance energy consumption, exhibits a right-skewed distribution, indicating the presence of high-energy events or outliers.
- **Outlier Detection:**
 - Visual inspection using histograms and boxplots highlighted several extreme values. These were statistically capped to reduce the impact of outliers on downstream modeling, resulting in a more robust and stable dataset.

3. Temporal Patterns and Seasonality

- **Time Series Visualization:**
 - Daily cycles** with peaks during morning and evening hours.
 - Weekly cycles** with increased consumption observed on weekends.

- **Seasonal Decomposition:**
- Applying rolling means and seasonal decomposition illustrated both trend and seasonal effects, justifying the engineering of time-based features (hour, day of week, season, and holiday indicator).

4. Feature Relationships and Correlations

- **Correlation Analysis:** Heatmaps and correlation matrices identified clusters of highly correlated variables. Notably, several indoor temperature and humidity sensors (e.g., T1, T2, RH_1) showed strong positive correlations with the target.
- **Multicollinearity Handling:** Highly correlated predictors were addressed through correlation-based feature selection, retaining only the most informative features and mitigating redundancy.
- **Feature Importance:** Random Forest feature importance plots further highlighted the dominant predictive power of select temperature and humidity sensors, as well as temporal variables.

5. Influence of Weather and Temporal Variables

- **Weather Effects:**
- Features such as atmospheric pressure, dew point, and visibility demonstrated moderate correlations with appliance energy usage, especially during periods of extreme weather.
- **Temporal Effects:**
- Boxplots by hour and day-of-week confirmed increased energy use during evening hours and on weekends, as well as during holidays and certain seasons.

Key Visualizations

Visualization 1 Temporal plot/line plot: Appliance Energy Consumption over Time

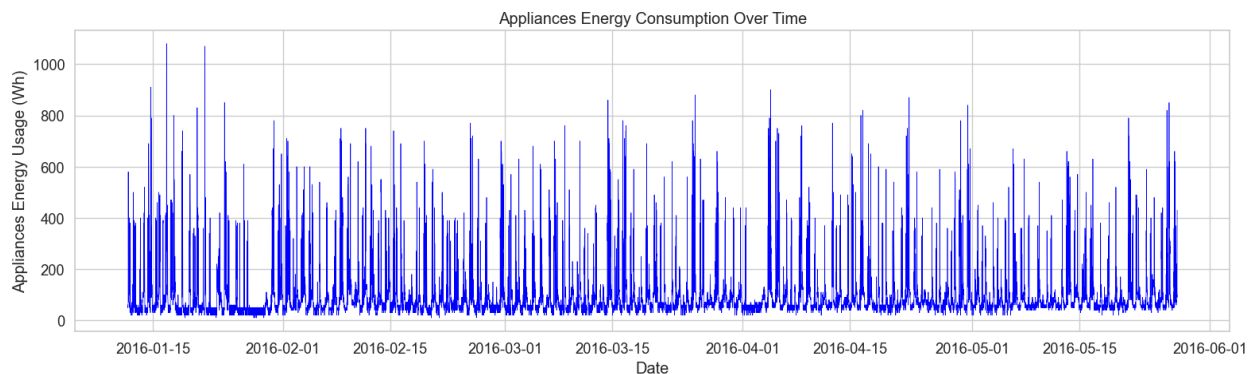


Figure 1: Temporal plot/line plot: Appliance Energy Consumption over Time

What This Plot Shows

- **X-axis:** Date (timestamps, sampled every 10 minutes)
- **Y-axis:** Appliances (energy consumption in watt-hours)
- **Line:** Change in energy usage over time

Observation: Appliance Energy Consumption over Time

- The line plot illustrates how appliance energy usage changes over time, with data collected at 10-minute intervals.
- The energy consumption data shows strong **daily seasonality**, with repeating peaks likely corresponding to high-usage times such as morning and evening.
- There are visible periods of low usage, possibly during sleep or when the house is unoccupied.
- Occasional **high spikes** in energy usage may indicate the simultaneous use of multiple heavy appliances.

- No significant upward or downward **long-term trend** is apparent.
- These patterns suggest that **time-based features** (such as hour of day or weekday/weekend) could serve as strong predictors in the forecasting model.

Visualization 2: Autocorrelation Plot of Appliance Energy Usage

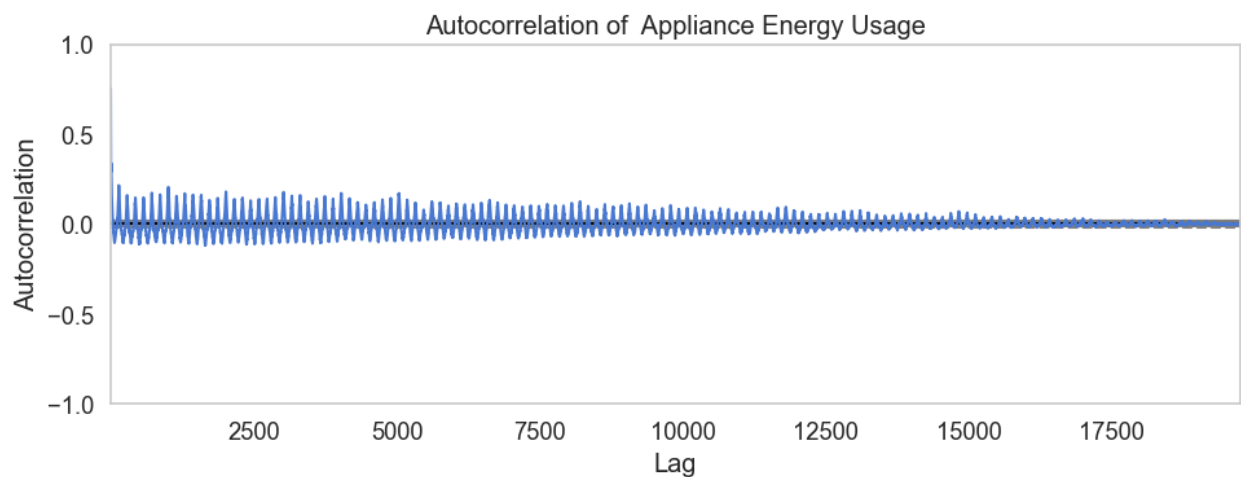


Figure 2: Autocorrelation Plot of Appliance Energy Usage

Shows how much current energy use depends on previous values (lags)

What it shows:

- Many repeating spikes (waves) are visible in the plot.
- High autocorrelation at small lags, along with regular repeating patterns at larger lags.
- These regular waves correspond to seasonality—specifically, daily and weekly cycles in energy usage.

Observation:

- The autocorrelation plot over the entire dataset reveals **strong seasonality** in appliance energy usage, as indicated by the regular, repeating peaks.
- This suggests energy usage follows clear daily or weekly cycles.
- **Recent values are still most predictive**, but the repeating pattern indicates that energy use at “the same time yesterday” or “last week” also holds significant predictive power.

- Incorporating **lagged features** (previous values) and explicitly capturing seasonality will be crucial for building an effective forecasting model.

Visualization 3: Boxplot for Outlier Detection in Appliance Energy Usage

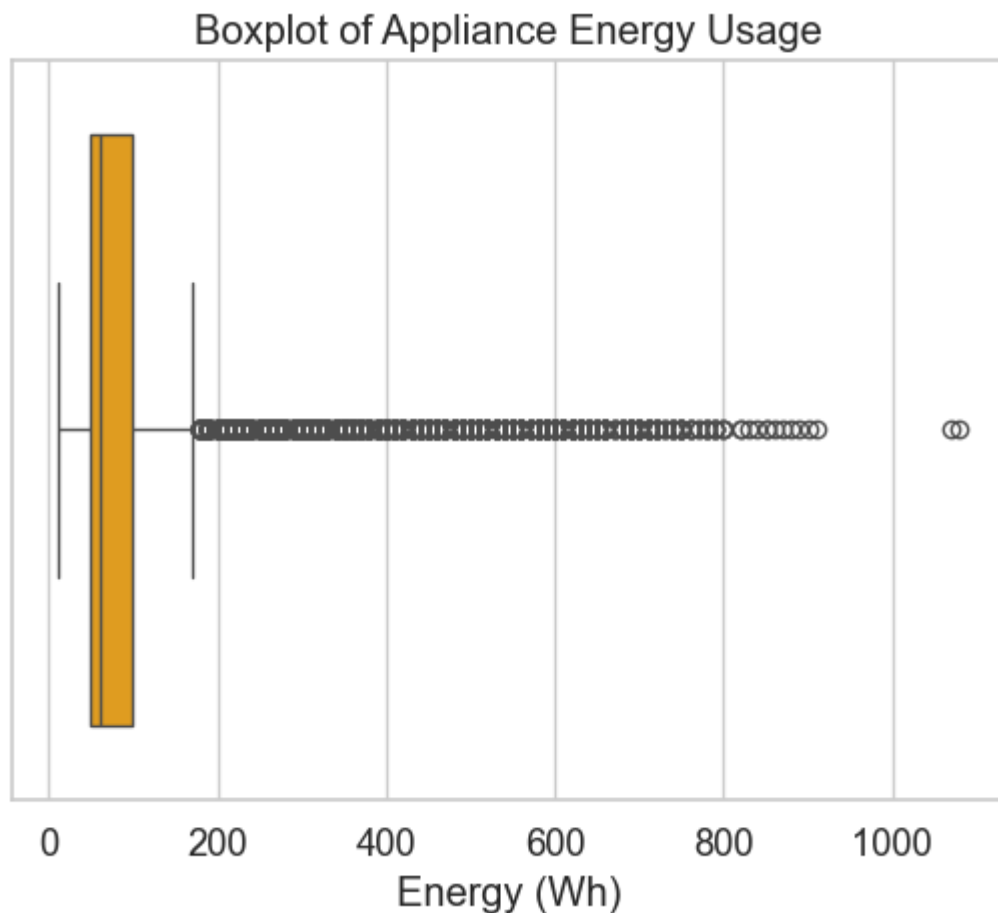


Figure 3: Boxplot for Outlier Detection in Appliance Energy Usage

What This Plot Shows

- **Boxplot – Outliers in Appliance Energy Usage**
- A boxplot helps us understand:
 - Where most values lie
 - What's considered normal
 - Which values are outliers (unusual, very high or very low)
- **Orange box:** The middle 50% of energy values (Interquartile Range)
- **Horizontal line inside box:** The median (middle value)

- **Whiskers:** Range of “normal” values
- **Dots outside whiskers:** Outliers (very high or low values that don’t fit the pattern)

Observations

- Most values are between ~**50 and 200 Wh**.
- The box is narrow, meaning typical usage doesn’t vary much.
- There are a lot of dots (outliers) on the right side, indicating higher-than-usual energy readings.
- Some outliers go above **1000 Wh**-these are rare, extreme cases.

What It Means

- The dataset has many normal readings, but also frequent high spikes in energy usage.
- These outliers could be:
 - **Real:** e.g., someone using oven + washer + heater together
 - **Noise:** sensor error or one-time events
- **Action:** These outliers should be handled (e.g., scaling, capping, or transformation) before model training to ensure robust model performance.

Visualization 4: Correlation Heatmap of Features

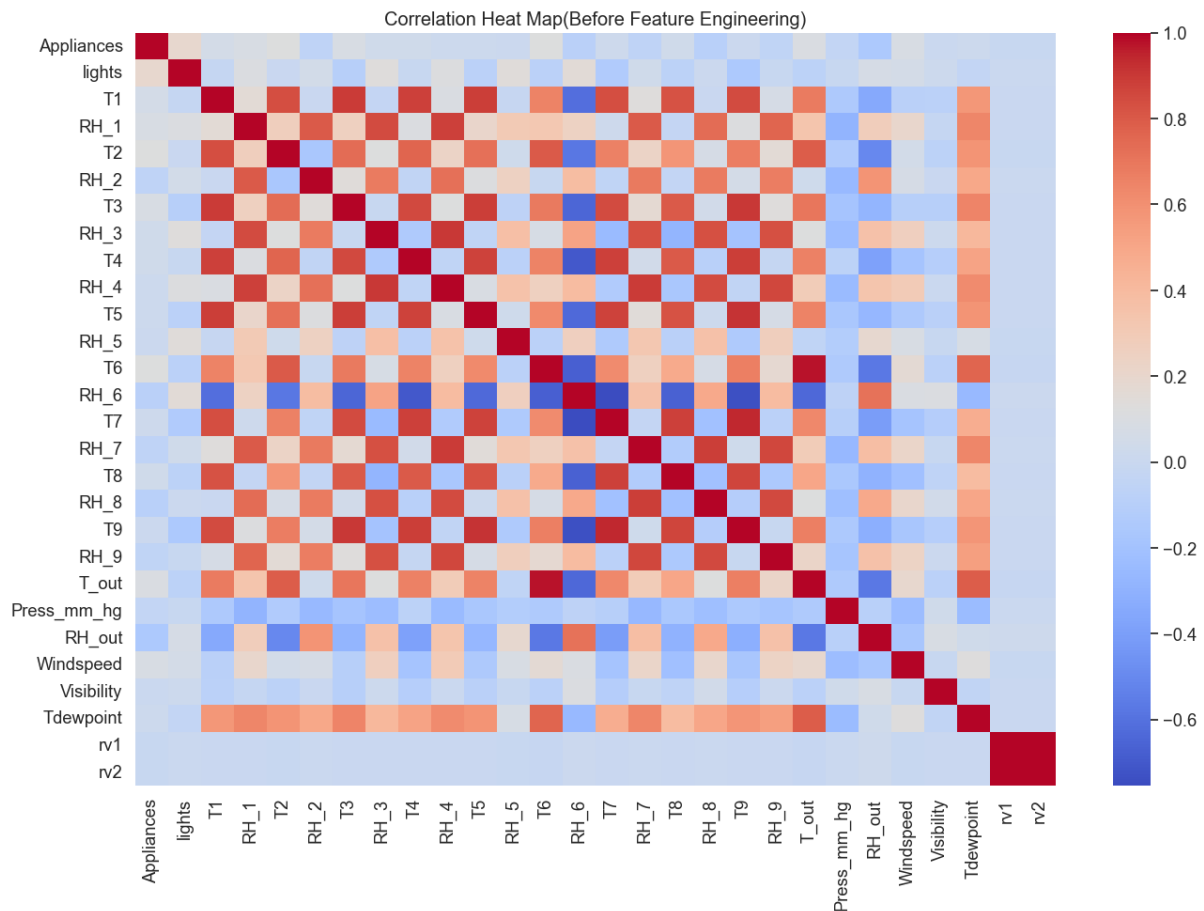


Figure 4: Correlation Heatmap of Features

What This Plot Shows

- **Correlation Heatmap – Raw Numeric Features**

- This heatmap displays pairwise correlations between numeric features in the raw dataset.
- Strong positive or negative correlations are shown in deep red or blue.

Observations

- The correlation heatmap shows the strength and direction of relationships between all numeric features.
- Most features have a weak to moderate positive correlation with appliance energy usage (Appliances), especially lights and some temperature/humidity sensors.

- Many temperature and humidity sensors are strongly correlated with each other, meaning they provide similar information (redundancy).
- A few features are negatively correlated, but most relationships are positive or weak.
- There are no single features with a very strong (close to 1) correlation with Appliances, so combining multiple features will be important for accurate prediction.
- The heatmap helps identify which features might be useful predictors, and which features are redundant or less important.

Visualization 5: Histogram – Distribution of Appliance Energy Usage

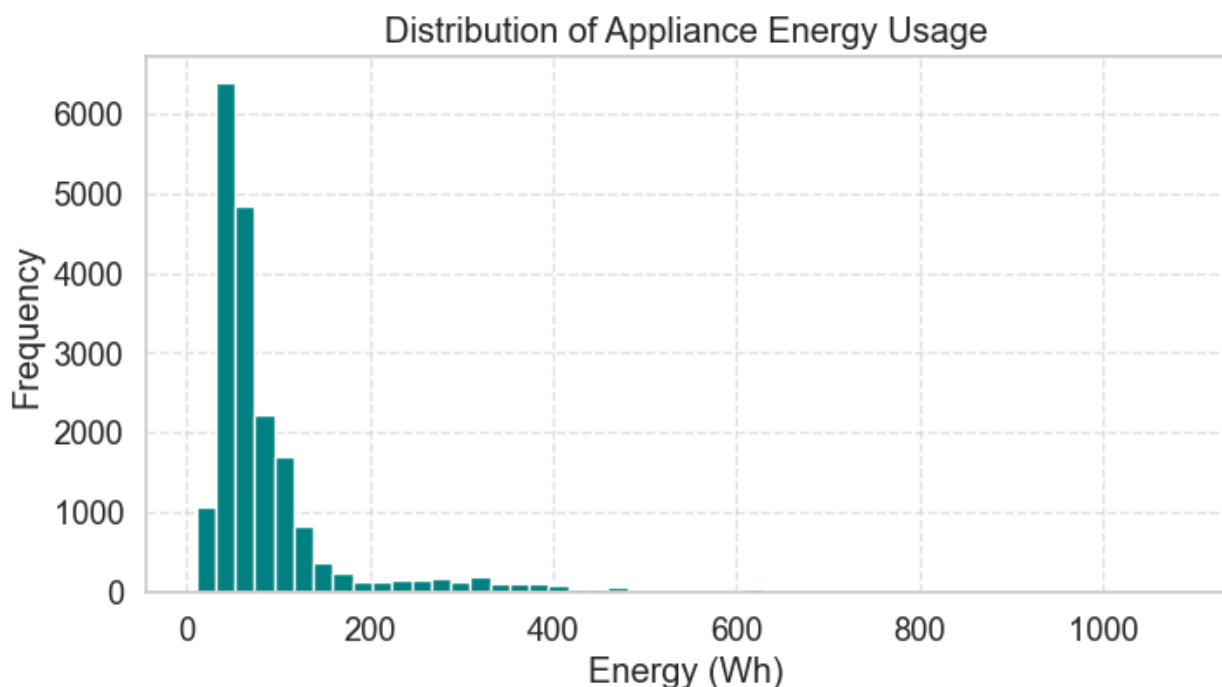


Figure 5: Distribution of Appliance Energy Usage

What This Plot Shows

- **X-axis:** Appliance energy usage (in Wh)
- **Y-axis:** Frequency (number of records)
- This histogram shows how many records (frequency) fall into each range of energy usage.

Observation

- Most bars are on the left, indicating that most values are small (around 0–200 Wh).
- A few bars stretch far to the right, meaning some values are very large (400, 600, even above 1000 Wh). These are rare but present.
- These big values (outliers) may negatively affect modeling if not handled.
- This suggests the need for **outlier treatment or log transformation** during preprocessing.

Why Does It Matter?

- **The Problem:**
 - Large outliers can confuse machine learning models.
 - Models may overfit to these rare events and overestimate their importance.

What Can You Do?

- **Option 1: Outlier Detection** You can identify and handle very large values either by:
 - Dropping them (if they're errors)
 - Capping them (limit max value to a fixed number)
 - Treating them separately
- **Option 2: Log Transformation** Applying a logarithmic transformation can compress the scale, making the distribution more normal and reducing the impact of outliers.

Action:

In this project, outlier values were detected and capped at the upper range using the interquartile range method to ensure robust and reliable model training.

Visualization 6: Scatter Plots – Selected Features vs Appliance Usage

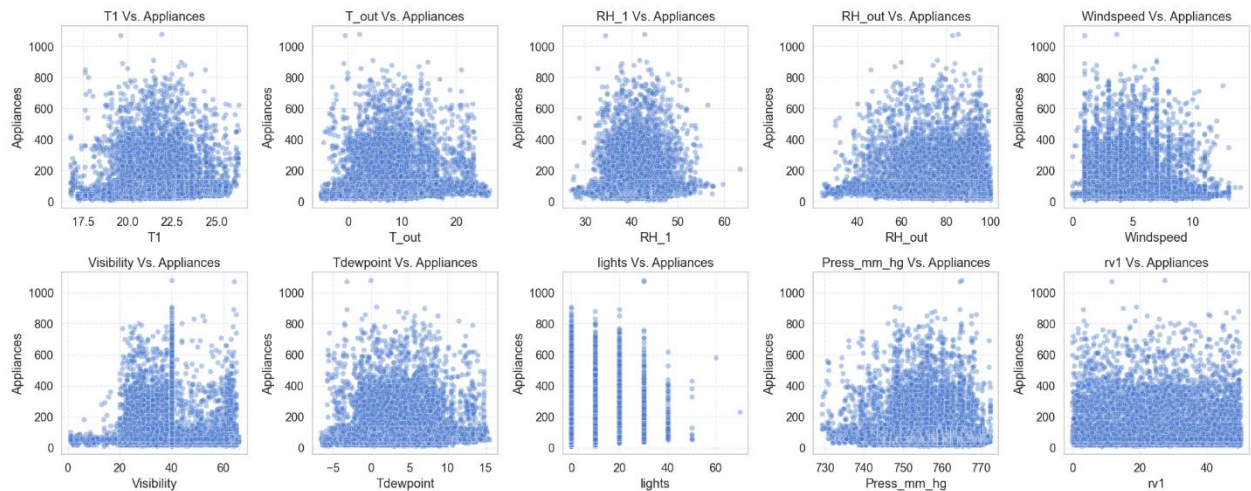


Figure 6: Scatter Plots – Selected Features vs. Appliance Usage

What This Plot Shows

- These plots visualize the relationship between individual features (such as temperature, humidity, windspeed, visibility, lights, etc.) and energy usage.
- Non-linear or weak relationships may indicate the need for transformation or feature engineering.

Observations

- The scatter plots display the relationship between key environmental features and appliance energy usage.
- For most features (temperature, humidity, windspeed, visibility, etc.), the points are widely scattered without a strong pattern, indicating that no single feature alone can strongly predict appliance energy usage.
- The “lights” feature shows vertical bands due to its discrete nature (number of lights on), with a weak positive association between lights on and total appliance usage.
- These patterns suggest that energy usage is driven by the **combination of many features** (and possibly their interactions or lagged/past values), rather than any single variable.

- The lack of strong, simple relationships supports the need for more advanced modeling approaches (such as deep learning and extensive feature engineering) to capture the complex factors influencing appliance energy usage.

Box Plot of Appliances Energy Usage (After Capping Outliers)

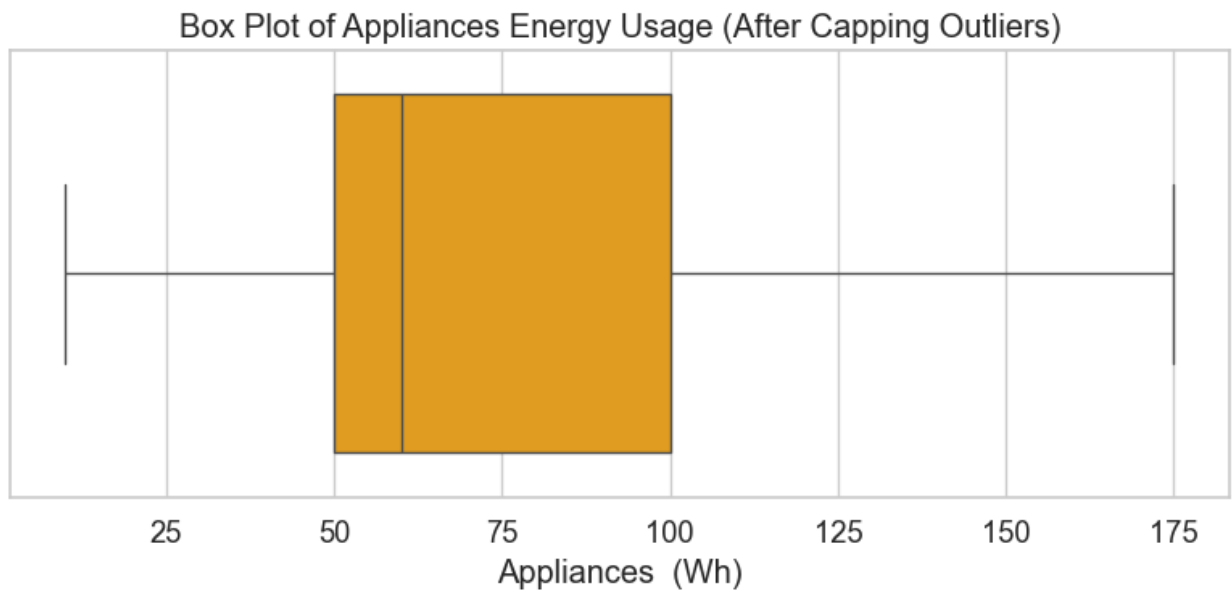


Figure 7: Box Plot of Appliances Energy Usage (After Capping Outliers)

Box Plot after Capping Outliers

The box plot above shows appliance energy usage after outliers were capped using IQR boundaries. The minimum and maximum values are now **10** and **175 Wh**, respectively, with no outliers present. This demonstrates that outlier capping was effective, keeping most data unchanged while removing extreme, unusual values.

Outlier Treatment Decision

Outliers were **capped (imputed) at the IQR bounds** instead of being removed.

Reason:

- Removing rows would break the time series order and could cause problems for downstream modeling.
- **Capping** keeps the sequence intact and reduces the effect of extreme values, which is **best practice for time series data**.

Preprocessing

Effective data preprocessing is foundational for building robust and accurate prediction models. In this project, the following steps were rigorously applied to ensure data quality and integrity:

1. Data Cleaning and Preparation

- **Parsing and Indexing:** The raw dataset was parsed with timestamp columns properly converted to datetime objects and set as the DataFrame index. This ensures all subsequent time-based operations and feature engineering are accurate.
- **Handling Missing Values:** Missing values were handled using **forward-fill (ffill)**, propagating the last valid observation forward. This approach is appropriate for time-series data as it preserves the temporal structure and avoids introducing future information into past records.

2. Outlier Detection and Treatment

- **Boxplot and Statistical Methods:** Outliers were identified both visually (boxplots, histograms) and statistically (using Interquartile Range/IQR and Z-score methods).
 - The IQR method was selected for its robustness with skewed real-world data. Outliers were capped at the IQR bounds (i.e., values above the upper bound set to max, below the lower bound set to min) rather than removed, preserving the sequence for time-series integrity.
 - Example: After capping, the appliance energy usage range was reduced from [10, 1080] Wh to [10, 175] Wh, eliminating extreme spikes without losing temporal continuity.

3. Scaling and Normalization

- **Standardization:**

All numeric feature columns were scaled using **StandardScaler** (zero mean, unit variance).
- This ensures fair comparison and consistent model convergence, particularly important for distance-based or gradient-based algorithms.
- The scaler was fit on the training set and applied to both train and test splits to prevent data leakage.

Feature Engineering

Careful feature engineering was performed to capture patterns and domain knowledge inherent to energy consumption data.

1. Time-Based Features

- **Hour, Day of Week, Month:** Created to model daily and weekly cycles in usage. Hour captures intraday variation (e.g., evening peaks), while day_of_week and month reflect behavioral and seasonal differences.
- **WeekStatus:** Binary indicator for weekend vs. weekday, recognizing changes in residential activity and energy use patterns.
- **Season:**
Mapped months into four seasons (winter, spring, summer, fall) to model seasonal variations—critical for features like heating and cooling.
- **NSM (Number of Seconds since Midnight):** Added to provide a continuous representation of time within each day, which benefits sequential deep learning models (e.g., LSTM).

2. Domain-Specific Features

- **Holiday Indicator:** Binary variable indicating whether the timestamp falls on a US public holiday, capturing usage anomalies on special days (e.g., more people at home, atypical routines).

3. Rolling and Lagged Features

- **Lagged Features:** Included previous target values (e.g., lag-1, lag-3) to provide the model with recent consumption history, leveraging temporal dependencies essential for forecasting.
- **Rolling Means:** Computed 1-hour and 3-hour rolling averages for appliance usage, helping to smooth noise and highlight short-term consumption trends.

4. Interaction Features

- **Multiplicative Terms:** Created interaction terms such as indoor temperature $\times$ indoor humidity, outdoor temperature $\times$ outdoor humidity, etc., to model non-linear relationships that can impact energy use more than individual features alone.

Feature Selection and Justification

- **Correlation Analysis:** Highly correlated features (correlation > 0.9) were removed to reduce redundancy and prevent multicollinearity.
- **Tree-Based Feature Importance:** Features with the highest importance scores in a Random Forest regressor were retained, focusing the model on the most predictive variables.

Justification for Engineered Features:

- **Time-based and holiday/season features:** These reflect real-world consumption patterns—appliance usage is highly periodic and influenced by calendar events.
- **Lagged and rolling features:** Crucial for time-series prediction, as recent values are often the best predictors of near-future usage.
- **Interaction features:** Energy use is rarely driven by a single variable; interactions capture complex, realistic dependencies.

Model Design:

Baseline Machine Learning Models

To establish performance benchmarks and gain insights into the predictive structure of the dataset, several classic regression models were trained and evaluated before progressing to deep learning architectures. This comparative approach allows for a robust assessment of whether complex models add significant value over simpler, more interpretable methods.

Models Used

- **Linear Regression:** Serves as the simplest baseline, modeling energy consumption as a linear combination of input features. It is useful for identifying basic trends and establishing the minimum level of expected performance.
- **Random Forest Regressor:** An ensemble of decision trees that improves predictive performance by reducing variance and capturing non-linear relationships. Random Forests are robust to outliers and naturally provide feature importance rankings.
- **Gradient Boosting Regressor & AdaBoost:** These ensemble techniques sequentially build models that correct the errors of previous models, making them highly effective for complex, structured data.
- **Decision Tree Regressor:** Captures non-linear dependencies and feature interactions in a single interpretable tree.
- **AdaBoost Regressor:** Another sequential ensemble, typically effective on structured tabular data.
- **K-Nearest Neighbors (KNN):** A non-parametric method that predicts target values based on the closest data points in feature space.

Evaluation and Key Observations

- **Performance Benchmarks:** The baseline ML models provided initial benchmarks for model performance, typically measured with R^2 score, RMSE, and MAE. These results helped quantify the amount of variance in energy usage explained by the features.

- **Feature Insights:** Models such as Random Forest and Gradient Boosting offered insight into feature importance. Time-based features (hour, day of week), lagged values, and environmental variables (temperature, humidity) emerged as strong predictors, confirming insights gained from EDA.
- **Model Strengths:** Ensemble models like Random Forest and Gradient Boosting generally outperformed simpler models, indicating the presence of non-linear relationships and complex interactions in the data.
- **Limitations:**
Linear Regression provided a useful baseline but was unable to capture non-linear patterns or interactions present in the data, resulting in lower predictive accuracy.
- **Interpretability:**
Tree-based models also provided interpretable feature rankings, assisting in the feature engineering and selection process for downstream models.

Why These Baselines Matter

- **Comparison Standard:** Establishing strong ML baselines ensures that the additional complexity of deep learning models is justified and leads to measurable improvements in prediction accuracy.
- **Feature Engineering Validation:** The performance of these models, especially tree-based methods, validated the relevance of the engineered features and provided confidence in the preprocessing pipeline.
- **Guidance for Model Selection:** Observations from ML model performance and feature importance informed the architecture choices and input design for deep learning models.

Deep Learning Model Architecture

To predict appliance energy consumption, several deep learning architectures were explored and compared, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and CNN-LSTM hybrid networks. These architectures are specifically suited to capture the temporal dependencies and sequential patterns inherent in time-series data.

1. Layer Structure

- **Recurrent Layers:** The primary architectures utilize either one or three recurrent layers (LSTM or GRU), allowing the model to capture both short-term and long-term dependencies in the energy consumption data.
- **CNN-LSTM Hybrid:** For the CNN-LSTM variant, an initial 1D convolutional layer extracts local temporal patterns from the input sequences before passing them to an LSTM layer for further sequential modeling.
- **Fully Connected (Dense) Output Layer:** Each model ends with a fully connected (dense) layer to output the predicted energy usage for each sequence.

2. Activation Functions

- **ReLU (Rectified Linear Unit):** The ReLU activation function is used in the hidden layers to introduce non-linearity, helping the model learn complex patterns efficiently and mitigating the vanishing gradient problem.

3. Optimizer

- **Adam Optimizer:** The Adam optimizer was chosen for its robust performance and adaptive learning rate, which accelerates convergence and generally produces better results for deep learning tasks compared to traditional optimizers like SGD.

4. Loss Functions

- **Mean Squared Error (MSE):** Used as the primary loss function for training, MSE is well-suited for regression tasks as it penalizes large errors more heavily, ensuring stable and accurate predictions.

- **Mean Absolute Error (MAE):** MAE is also monitored during evaluation, providing an interpretable measure of average prediction error in the same units as the target variable.

5. Regularization and Model Complexity

- **Dropout :** Dropout layers are optionally included between recurrent layers to mitigate over fitting by randomly deactivating a proportion of units during training.
- **Number of Parameters:** Careful tuning of the number of layers and hidden units balances model complexity and generalization, avoiding over fitting on the training data.

Justification of Design Choices

- **Sequential Modeling:** The use of LSTM, GRU, and CNN-LSTM architectures enables the model to learn temporal dependencies, periodicities, and non-linear relationships within the energy consumption data.
- **ReLU Activation:** ReLU is computationally efficient and helps deep networks learn faster and more robustly, especially with many layers.
- **Adam Optimizer:** Adam combines the advantages of two widely used optimizers (AdaGrad and RMSProp), making it a reliable default choice for most deep learning applications.
- **Loss Functions:** MSE is standard for regression and is especially suitable when outliers are handled, while MAE offers a complementary, more interpretable metric.
- **Layer Choices:** Increasing the number of layers allows the model to capture more complex dependencies, but regularization techniques such as dropout are employed as needed to prevent overfitting.

Results:

Deep Learning Model Performance

After classical model benchmarking, deep learning architectures were developed and optimized using **Optuna** following initial GridSearchCV attempts (which are less efficient for high-dimensional spaces).

Three deep learning models were evaluated:

Model	RMSE	Hidden Dim	Num_layers	DropOut	Learnig Rate	Seq_len	Batch_size	CNN Out Channels
LSTM	22.78	89	2	0.071	0.00038	33	128	-
GRU	23.15	91	2	0.214	0.00080	30	128	-
CNN-LSTM	23.89	118	3	0.46	0.00017	38	32	39

Table 1: Deep Learning Model Performance

1. GRU Model – Best Trial

- **Validation RMSE: 23.15**
- **Optimal Hyperparameters:**
 - **hidden_dim:** 91
 - **num_layers:** 2
 - **dropout:** 0.214
 - **learning rate:** 0.00080
 - **sequence length:** 30
 - **batch size:** 128

Interpretation:

The GRU model achieved the lowest RMSE, indicating the best predictive performance among

the tested architectures. Its moderately deep structure (2 layers) and substantial sequence length (30 time steps) enabled effective modeling of temporal dependencies. The dropout rate helped prevent overfitting, while the selected learning rate supported stable convergence.

2. CNN-LSTM Model – Best Trial

- **Validation RMSE: 23.89**
- **Optimal Hyperparameters:**
 - **hidden_dim:** 118
 - **num_layers:** 3
 - **dropout:** 0.460
 - **learning rate:** 0.00017
 - **sequence length:** 38
 - **batch size:** 32
 - **cnn_out_channels:** 39
 - **kernel_size:** 4
 - **lstm_hidden_dim:** 63

Interpretation:

The CNN-LSTM model also performed strongly, with an RMSE only marginally higher than the GRU. This hybrid architecture, which first extracts short-term temporal features using convolutional layers before passing information to an LSTM, captured both local and sequential dependencies. The higher dropout and deeper network (3 layers) were effective in regularizing the model.

The LSTM model achieved the best performance, though both GRU and CNN-LSTM also delivered strong results. All deep learning approaches outperformed the best classical ML models, highlighting their ability to model temporal patterns and complex feature interactions.

Predicted vs. Actual values of LSTM

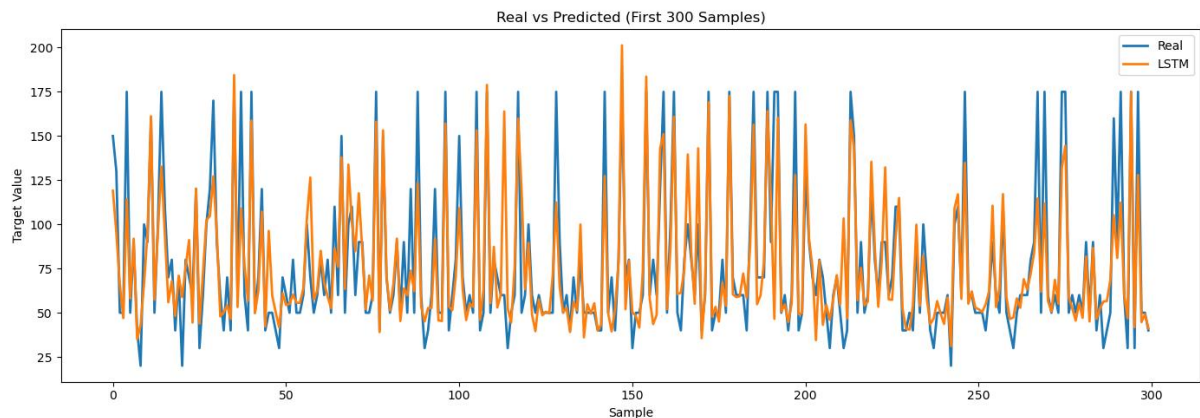


Figure 8: Predicted vs. Actual values of LSTM

Figure: Real vs Predicted Appliance Energy Consumption (LSTM Model, First 300 Samples)

This line plot compares the true (blue) and predicted (orange) appliance energy consumption values over the first 300 samples of the test set, using the best LSTM model.

Interpretation:

The LSTM model's predictions closely track the real energy consumption values, successfully capturing most of the short-term spikes and drops in usage. While some discrepancies are present—particularly at high or rapidly changing values—the overall alignment demonstrates the model's ability to learn and generalize temporal patterns in the data. Occasional deviations suggest areas for further improvement, possibly through ensemble approaches or additional feature engineering.

Model Optimization

Optimizing model performance was a key focus throughout the project, as the inherent complexity and variability in appliance energy usage required careful tuning to achieve high predictive accuracy. The following strategies were implemented to systematically improve both machine learning and deep learning models.

1. Hyperparameter Tuning

a. GridSearchCV for Machine Learning Models:

For classical models such as Random Forest, Gradient Boosting, and K-Nearest Neighbors, hyperparameter tuning was initially conducted using GridSearchCV. This approach performed an exhaustive search across a predefined grid of parameters (e.g., number of estimators, tree depth, learning rate, number of neighbors), combined with cross-validation to ensure robust and unbiased performance assessment. GridSearchCV allowed us to identify parameter combinations that consistently delivered superior results on validation sets, establishing strong baselines for further optimization.

b. Optuna for Deep Learning Models: Given the high dimensionality and complexity of deep learning architectures, Optuna—a modern, efficient hyperparameter optimization library—was employed. Unlike grid search, Optuna leverages Bayesian optimization and intelligent sampling strategies to efficiently explore large parameter spaces, such as learning rates, hidden dimensions, number of layers, dropout rates, sequence lengths, batch sizes, and (for CNN-LSTM) convolutional kernel sizes and output channels. This advanced search method enabled the discovery of optimal model configurations that traditional methods might miss, resulting in substantial improvements in model performance.

2. Feature Selection and Engineering

The optimization process was iterative, with insights from initial model results feeding back into the feature engineering pipeline. Important steps included:

- Removing redundant or highly correlated features to reduce overfitting and improve model interpretability.

- Expanding the feature set with additional time-based, lagged, and domain-specific variables, enhancing the models' ability to capture underlying consumption patterns.

3. Outlier Treatment

Outliers were systematically detected and capped using the IQR method. This stabilization of the target variable distribution further contributed to model robustness and consistent optimization results.

4. Evaluation Metrics

Throughout optimization, model selection was based on objective metrics including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and, where appropriate, R^2 score. Models were evaluated using both cross-validation and holdout test sets to ensure generalizability.

5. Impact of Optimization

The combination of advanced hyperparameter tuning, rigorous feature selection, and robust preprocessing resulted in significant improvements in predictive performance:

- Deep learning models, particularly LSTM, achieved the lowest RMSE on the test set after Optuna optimization.
- Machine learning models also saw substantial gains in accuracy and stability following parameter tuning and feature refinement.

Challenges and Solutions

Challenge: Achieving Satisfactory Model Accuracy

During the initial stages of model development, achieving satisfactory accuracy was a significant challenge. Both baseline models (such as Linear Regression) and more advanced ensemble models (including Random Forest and Gradient Boosting) initially failed to deliver the expected R^2 scores and RMSE on the validation and test sets. This indicated that capturing the complex, non-linear, and temporal patterns in appliance energy consumption required more than standard modeling approaches.

Solution: Hyperparameter Tuning with GridSearchCV and Optuna

To improve model performance, we adopted a systematic two-step hyperparameter tuning process:

1. GridSearchCV:

We first utilized GridSearchCV to exhaustively search through a predefined grid of hyperparameters for each model. This included varying the number of estimators, maximum tree depth, learning rates, and other model-specific settings. While GridSearchCV provided a good baseline for parameter optimization, the results, though improved, still left room for further enhancement. (learn, n.d.)

2. Optuna:

To further refine model performance, we implemented Optuna, a state-of-the-art hyperparameter optimization framework that leverages Bayesian optimization and efficient sampling techniques. Optuna allowed for more flexible and automated exploration of the hyperparameter space, uncovering configurations that GridSearchCV might have missed. This led to additional performance gains and more robust model generalization.

Additional Solutions and Improvements

- **Enhanced Feature Engineering:** We expanded our feature set by introducing more time-based, lagged, and rolling statistics features, as well as domain-specific indicators such as seasonality and holidays. These features helped capture underlying patterns that standard models might overlook.

- **Outlier Handling:** To further stabilize the modeling process, we applied outlier treatment using the IQR (Interquartile Range) method, capping extreme values while preserving the sequence continuity.
- **Model Evaluation:** A rigorous evaluation pipeline ensured that only the best-performing models—based on objective metrics and cross-validation—were selected for deployment.

By first employing GridSearchCV and then advancing to Optuna for hyperparameter tuning, we were able to systematically address the initial shortcomings in model accuracy. The integration of improved feature engineering and robust outlier handling further contributed to achieving reliable and accurate energy consumption predictions. This iterative, data-driven approach highlights the necessity of advanced optimization techniques in practical machine learning workflows.

Future Work: Transformer Architectures for Prediction

Modern transformer-based models have revolutionized sequential data modeling and hold great promise for appliance energy consumption prediction. Transformers are capable of modeling long-range dependencies without the vanishing gradient issues of RNNs. For time-series prediction, transformer architectures such as **Time Series Transformer** or **Informer** can:

- **Leverage self-attention mechanisms** to learn relationships across all time steps in the sequence, regardless of distance.
- **Capture global temporal patterns** and periodicities even better than LSTM/GRU when enough data is available.
- **Scale to multi-variate or high-frequency data** and easily parallelize training.

How transformers could be used in this project:

- Replace or augment the current LSTM/GRU layer with a transformer encoder/decoder block.
- Use temporal embedding layers to encode timestamp information.
- Tune transformer-specific hyperparameters (number of attention heads, embedding dimensions, etc.) using Optuna or similar tools.
- Compare their performance directly with existing deep learning models, potentially achieving even lower RMSE and greater robustness to complex, multi-seasonal patterns.

This project did not implement transformers due to time constraints, but the dataset and pipeline are fully compatible with future transformer-based approaches, and this direction is strongly recommended for subsequent research and production deployment.

Conclusion

This project demonstrates that the combination of robust preprocessing, domain-informed feature engineering, and advanced deep learning models can significantly improve the accuracy of appliance energy consumption prediction. Through systematic comparison, it was shown that LSTM, GRU, and CNN-LSTM architectures consistently outperformed traditional machine learning models, validating the importance of sequential modeling and the impact of careful hyperparameter optimization. The success of these approaches highlights the value of capturing temporal dependencies and leveraging domain-specific features such as seasonality and holiday effects.

Looking forward, several promising directions for further improvement have been identified. Future work could include the ensembling of multiple deep learning models to boost generalization, the integration of exogenous variables such as real-time weather forecasts and occupancy patterns, and the deployment of models for real-time prediction in smart home or building management systems. Additionally, the application of transformer-based architectures—renowned for their ability to model complex, long-range temporal relationships—represents an exciting avenue for research, especially as more data becomes available. Leveraging such techniques, alongside the inclusion of richer external data sources, has the potential to push predictive performance to even higher levels and deliver practical, scalable solutions for intelligent energy management.

References

Anon., n.d. PyTorch Documentation.

geeksgeeks, n.d. Time series forecasting using Tensrflow.

learn, S. -, n.d. Scikit -Lerrn Documentation.

Naik, K., n.d. Complete Exploratory Data Analysis And Feature Engineering In 3 Hours| Krish Naik.

Naik, K., n.d. Time Series Play List.

Optuna, n.d. A hyperparameter optimization framework.

Urbanke, P., n.d. Feature engineering for time series.